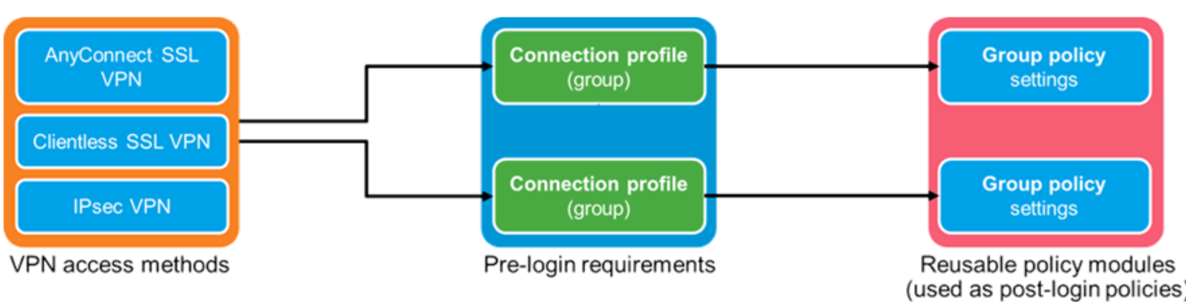
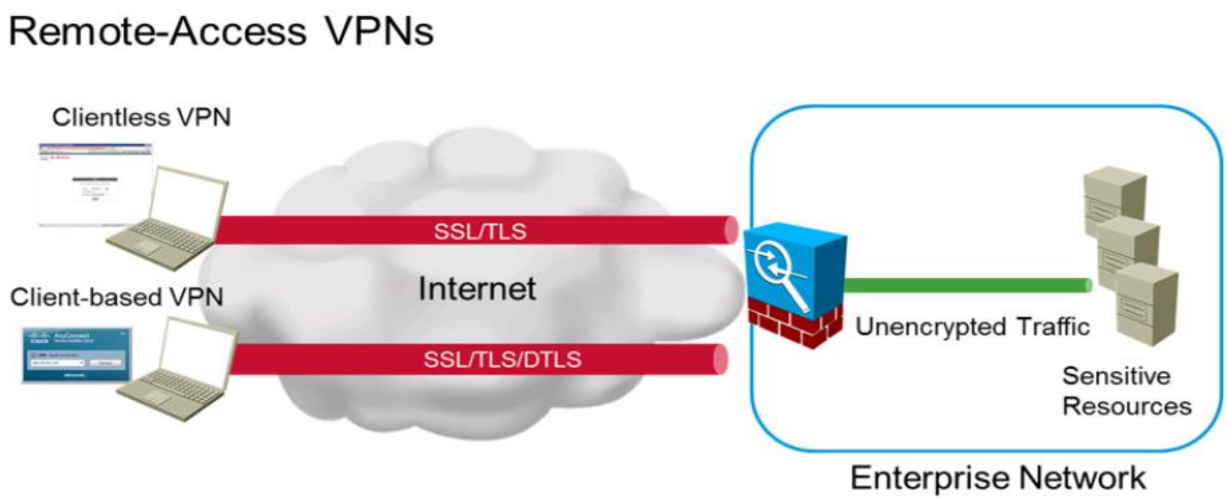
Remote-Access VPNs:

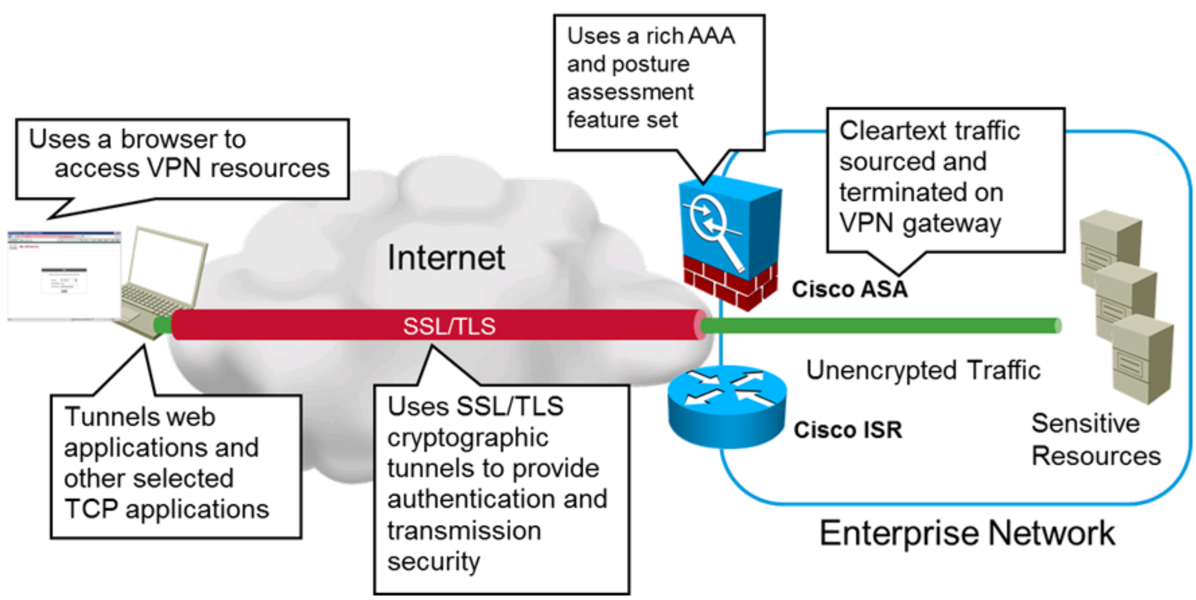
- VPN encryption prevents third parties from reading data as it passes through Internet.
- IPSec and SSL are the two most popular secure network protocol suites used in VPNs.
- IPSec and SSL are both designed to secure data in transit through encryption.
- IPsec VPNs and SSL VPNs both encrypt network data, but they do it differently.
- IPsec operates at network layer of OSI model and SSL operates at the transport layer.
- SSL is used to encrypt data sent between any two processes by port numbers.
- Internet protocol security defines official architecture for securing IP network traffic.
- IPsec specifies ways in which IP hosts can encrypt & authenticate data being sent.
- IPsec is used to create a secure tunnel between entities that are identified by their IP.
- IPsec VPNs typically are used to connect a remote host with a network VPN server;
- The traffic sent over the public internet is encrypted between VPN server & remote host.
- IPsec enable which cryptographic algorithms are to be used to encrypt or authenticate data.
- IPsec VPNs usually require each remote endpoint to use specific software to create tunnel.
- Modern SSL VPNs actually use TLS to encrypt streams of network data being sent.
- TLS protocol enables encryption & authentication of connections between programs.
- These connections are usually defined by the IP addresses of the endpoints.
- As well as the port numbers of the programs running on those endpoints.
- TLS enables communicating hosts to negotiate which cryptographic algorithms to use.
- Enable users to work from remote locations such as their homes & other premises.
- Remote-Access VPNs connect client devices to LAN over the Internet infrastructure.
- Individual hosts or clients access a company network securely over the Internet.
- Each host typically has VPN client software loaded or uses a Web-Based client.
- Whenever the host send any information, the VPN client software encapsulates it.
- Whenever the host send any information, the VPN client software also encrypts it.
- It allows individual users to establish secure connections with a remote network.
- Remote-Access VPN tunnels are formed between a VPN device & an end-user PC.
- The remote user requires the Cisco Virtual Private Network (VPN) client software.
- Remote access Virtual Private Network connect individual users to private networks.
- Remote-access Virtual Private Network connects individual host to company Network.

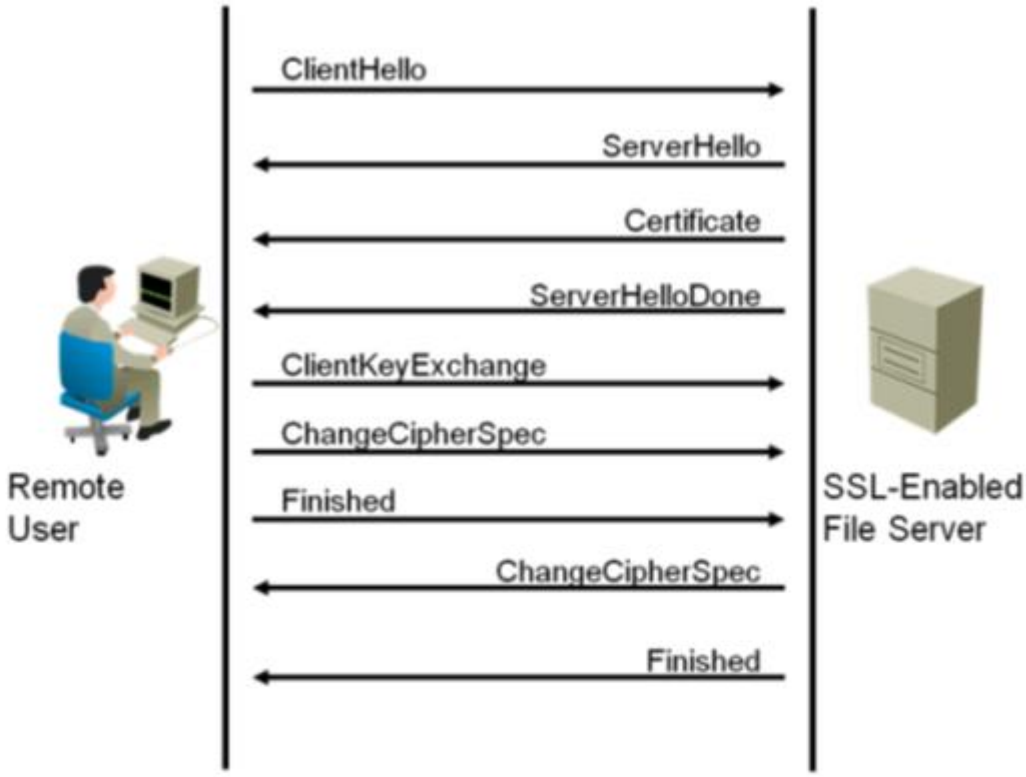




Clientless SSL VPN:

- Clientless SSL VPN enables end users to securely access resources.
- Clientless SSL VPN access corporate network from anywhere using Web browser.
- The user first authenticates with a Clientless SSL VPN gateway Firewall or Router.
- Which then allows the user to access pre-configured network resources.
- Clientless SSL VPN creates a secure, remote-access VPN tunnel to an ASA.
- It is using a Web Browser without requiring a software or hardware client.
- It provides secure and easy access to a broad range of Web resources.
- Connect to web page portal from which can have access to web-based applications.
- Clientless SSL VPN mode provides secure access to private web resources.
- You can access all the resources in your company which uses a web interface.
- Both ASA Firewall and Cisco IOS Routers support Web VPN technologies.





Client-Based VPN:

- Full-Tunnel mode provide access to virtually any application inside company.
- Client-based VPN is VPN created between a single user and a remote network.
- There's often an application involved to make the VPN connection.
- User manually starts VPN client & authenticates with username and password.
- Client creates encrypted tunnel between user's computer & remote network.
- The user then has access to the remote network via the encrypted tunnel.
- Examples of client-based VPN applications include Cisco's AnyConnect.
- VPN client software is installed on user's PC to provide remote access to network.
- Uses the IPSec protocol and provides full network connectivity to the remote user.